

HEMODYNAMICS OF DIASTOLIC FUNCTION IN HEALTH AND DISEASE

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Systolic function and measurement of the ejection fraction are often the primary focus of assessments of the health status of the left ventricle (LV). However, the physiologic events occurring during diastole are also important for optimal left ventricular function. The hemodynamics of left ventricular diastolic function in both health and disease states can be assessed by both invasive catheterization techniques and the more commonly used echocardiographic methods.

PHYSIOLOGIC EVENTS DURING DIASTOLE

There are four main stages of diastole in a normal heart (Fig. 12.1):

1. Isovolumetric relaxation

Diastole begins with this stage immediately after the aortic valve closes. The stage of isovolumetric relaxation ends just prior to the opening of the mitral valve. During this stage, the LV volume does not change, and the ventricular myocardium actively relaxes (an energy-dependent process) to its resting unstressed length. This causes a decrease in LV pressure. Once the LV pressure falls below the left atrial (LA) pressure, the mitral valve will open and the LV begins to fill, thus ending the stage of isovolumetric relaxation. This event can be seen on hemodynamic LV and LA waveform tracings (Fig. 12.2) just after the peak of the “v” wave when the “y” descent of the atrial tracing meets the descending slope of the LV pressure tracing. On echocardiography, this stage is described by the isovolumetric relaxation time (IVRT) and is typically 70 to 90 ms in those with normal diastolic function.¹

2. Rapid early ventricular filling

When the mitral valve opens, there is a pressure gradient between the LA and LV, created by the swift recoil of the LV with the rapid decline in pressure that occurred during isovolumetric relaxation, and blood is rapidly pulled through the mitral valve into the LV cavity. On hemodynamic waveforms, this phase is manifested by a sudden rise in the left ventricular diastolic pressure.

3. Diastasis

This stage occurs from mid-diastole to just before atrial contraction. The rate of filling slows as the LV diastolic pressure rises and reaches its peak. There may be continued filling if there is significantly elevated LA pressure or if the LV is compliant and allows additional filling.¹

4. Atrial contraction

Diastole is concluded when atrial contraction occurs and provides the final filling. This occurs immediately before the mitral valve closes. On the left ventricular tracing, this stage may be manifested by a prominent “a” wave, although it may not be observed in healthy compliant ventricles and may only be seen with stiffer noncompliant ventricles. On an echocardiogram, this stage produces the “A” wave on Doppler assessment.¹

ASSESSMENT OF DIASTOLIC FUNCTION BY CARDIAC CATHETERIZATION

Perhaps the simplest assessment of diastolic function is made during routine left heart catheterization with measurement of the left ventricular pressure (see Fig. 12.2). The left ventricular end-diastolic pressure (LVEDP) is elevated in both systolic and diastolic heart failure. The LVEDP is identified as the pressure just after the “a” wave and before the rise in systolic pressure that occurs with the onset of ventricular contraction (in line with the QRS complex) and is best measured at end expiration. Patients with abnormal diastolic function may also have an elevation of diastolic pressure early in diastole since the ventricle relaxes abnormally, thus elevating early diastolic pressure. This finding is commonly observed in patients with hypertrophic cardiomyopathy. For patients undergoing right heart catheterization, the pulmonary capillary wedge pressure (PCWP) reflects LA pressure. Patients with diastolic heart failure may manifest elevations in the PCWP with prominent “v” waves due to the fact that LA volume and pressure are already high, and further passive filling of the left atrium while the mitral valve is closed may result in marked pressure elevation and thus a prominent “v” wave. The PCWP correlates better with the mean LA pressure and pre-A wave pressure than with the LVEDP.^{2,3} Furthermore, the PCWP can be used for risk assessment and is a better prognostic marker for mortality in patients with heart failure with preserved ejection fraction (HFpEF) than the LVEDP.^{1,4,5}

Being able to assess diastology in the cardiac catheterization laboratory is a valuable tool that can offer essential prognostic information as well as diagnose the condition of HFpEF in cases where noninvasive hemodynamic assessment may have inadequate sensitivity.

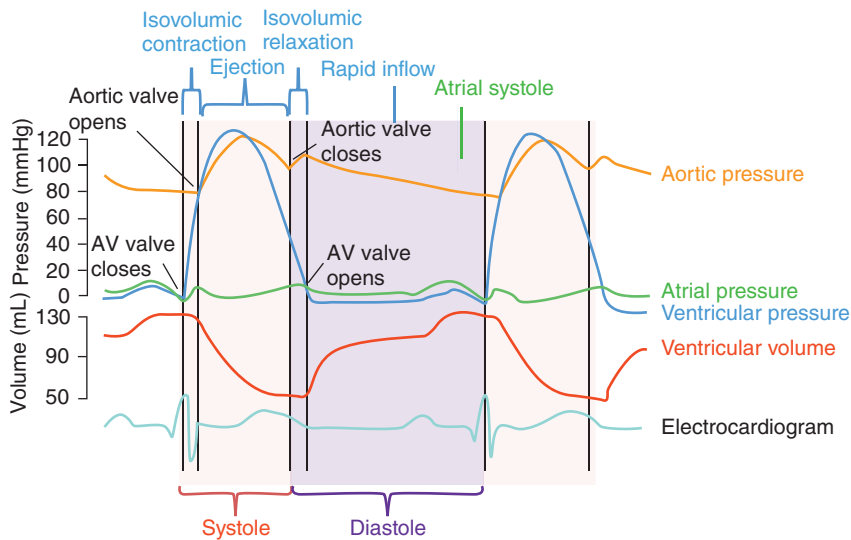


Fig. 12.1 Pressure and volume changes occurring within the cardiac chambers during the normal cardiac cycle. AV, Atrioventricular.

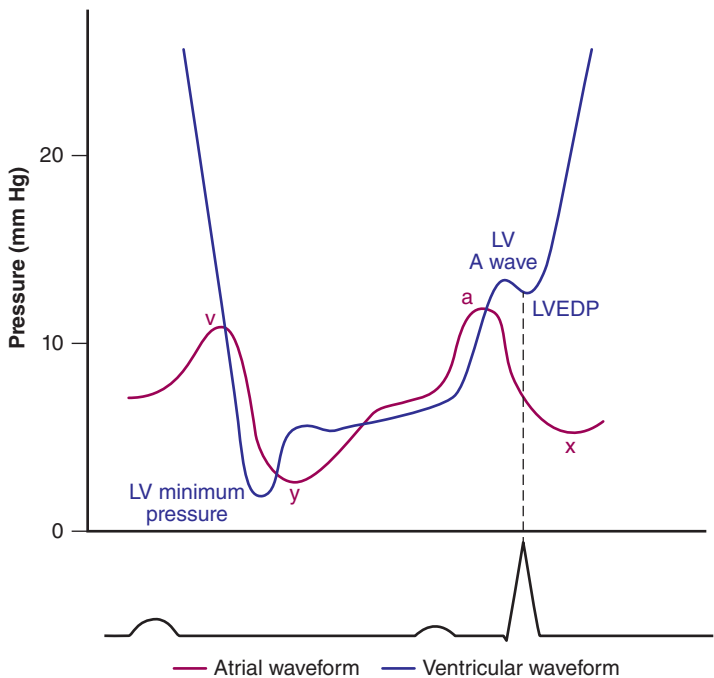


Fig. 12.2 Schematic representation of simultaneous left ventricular and left atrial pressures. LV, Left ventricular; LVEDP, left ventricular end-diastolic pressure.

More sophisticated invasive methods of assessing diastolic function have been developed and are extremely valuable for research applications but are not used clinically. These are based on pressure volume loops described in detail in Chapter 11 and include the measurement of ventricular compliance, relaxation rate, and LV diastolic time constant (Tau). Detailed information about these methods has been described elsewhere and is beyond the scope of this chapter.

ASSESSMENT OF DIASTOLIC FUNCTION BY ECHOCARDIOGRAPHY

In routine clinical practice, the diastolic function is primarily assessed noninvasively by echocardiographic techniques, in which multiple echo parameters and algorithms are used to determine either normal diastolic function in health state or diastolic dysfunction in disease state.

The echo parameters used include the E wave, A wave, E/A ratio, e' tissue velocities, E/ e' ratio, E wave deceleration time (DT), IVRT, pulmonary vein inflow velocities (S/D ratio), left atrial volume index (LAVI), and tricuspid regurgitation (TR) velocity. While it is important to become familiar with all parameters mentioned, the most important measurements clinically used to assess diastole and diagnose the presence and severity of dysfunction include the E wave, E/A ratio, E/ e' ratio, TR velocity, LAVI, and S/D ratio.

With normal diastolic function on echocardiography, there will typically be a combination of the following findings: an E/A ratio between 0.9 and 1.5 (unaffected by Valsalva maneuver), a DT between 140 and 240 ms, a TR velocity of <2.8 m/s, an LAVI of <34 mL/m², an average E/ e' of <14 , a septal e' velocity of >7 cm/s, and a lateral e' velocity of >10 cm/s.²

OVERVIEW OF ECHOCARDIOGRAPHIC PARAMETERS

There are several echocardiographic variables used to assess the LV diastolic function. The *mitral inflow velocity (E wave)* represents the LA-LV pressure gradient during early diastole (prior to atrial contraction) and is driven by left-sided filling pressures (LA pressure), early LV diastolic pressure, the rate of LV relaxation (Tau index), and the heart rate. A high LA pressure in the setting of diastolic dysfunction typically results in a large E wave; however, this may not always be the case. If LV relaxation is also severely abnormal (as seen in hypertrophic cardiomyopathy), then the early LA-LV gradient may be reduced, resulting in a lower E wave than expected despite diastolic dysfunction. Similar to high LA filling pressures, a high elastic recoil (as seen in the young population) can also result in a large E wave, but this does not necessarily mean that diastolic dysfunction is present.²

The A wave represents the mitral inflow velocity that occurs as a result of atrial contraction. The *mitral E/A ratio* is used to determine the LV filling pattern. It will help determine the presence and extent of diastolic dysfunction. These filling patterns include: (1) normal, (2) impaired relaxation, (3) pseudonormalization, and (4) restrictive. The filling pattern is classified as "indeterminate" if there is atrial fibrillation/atrial flutter, where atrial contraction is lost and the E/A ratio is unable to be calculated. Other limitations to its use may occur with sinus tachycardia, a prolonged PR interval, and a paced rhythm, where fusion of the E and A waves may occur.²

The *mitral annular recoil velocity (e')* is measured at both the septal and lateral mitral annulus via tissue Doppler on echo and is correlated with LV relaxation and restoring forces. As LV relaxation and restoring forces decrease with worsening diastolic dysfunction, the e' velocities will decrease as well. This value is typically reduced in both systolic and diastolic dysfunction and is independent of LV filling pressures (i.e., independent of the preload state). There is a significant association between this value and the Tau index measured in the catheterization laboratory.^{1,2,4}

We can use the *E/ e' ratio* to determine the left-sided filling pressure. The E/ e' value of >14 correlates with a significantly elevated filling pressure, and a value of <8 indicates a low filling pressure. When this ratio is between 8 and 14, other echocardiographic measurements and serum markers such as B-type natriuretic peptide should be used for further assessment of filling pressures. Clinical cases where the E/ e' ratio does not correlate with the left-sided filling pressures and may be falsely elevated include patients with mitral stenosis, a prosthetic mitral valve, a mitral ring, or severe mitral calcification. This can be explained by the fact that the e' values will be lower due to reduced motion of the mitral annulus rather than impaired LV relaxation.^{1,4}

The *LAVI* represents the left-sided filling pressures of the heart over time. An elevated LAVI suggests increased LA pressures over a significant amount of time.²

The *mitral E-velocity DT* represents the LV compliance and the rate at which the LV pressure rises once mitral inflow occurs during early diastole. It is influenced by LV stiffness, LV diastolic pressures as the mitral valve opens, and active LV relaxation (dP/dt or Tau index). This value typically increases with age. Patients with a shorter DT typically have an increased baseline LVEDP and decreased compliance. Normal DT is 140 to 240 ms. In mild forms of diastolic dysfunction (grade I), the DT is prolonged and usually >240 ms. In moderate diastolic dysfunction (grade II), there is "pseudonormalization" as the DT appears to return to the normal range (140–200 ms). In more severe grades of diastolic dysfunction (grade III or IV), the DT is usually shorter than normal at <140 ms. The shift from a prolonged DT in grade I dysfunction to a normalization and then shortening of the DT in grades II and III/IV dysfunction, respectively, is due to a continued loss of elastic recoil and progressive increase in the LA pressure. At these more advanced stages of diastolic dysfunction, the LV filling forces are generated predominantly by the elevated LA pressures pushing blood into the LV. As the LV cavity fills, its decreased compliance results in a rise in the LV filling pressure, and the LV pressure equilibrates with the LA pressure more rapidly. In patients with diastolic dysfunction, a DT of <130 ms is associated with a higher mortality.^{1,2}

The *IVRT* represents the duration of time between aortic valve closure and mitral valve opening during very early diastole. It is influenced by the rate of active LV relaxation (which decreases as diastolic dysfunction worsens) and by the left-sided filling pressures. As the rate of active LV relaxation decreases with grade I diastolic dysfunction, the *IVRT* increases. However, the significant increase in the left-sided pressures occurring during grades II to IV diastolic dysfunction offsets the effects of decreased LV relaxation, and the *IVRT* instead decreases significantly to values within the normal range for grade II dysfunction (i.e., “pseudonormalization”) and then values become lower than normal with more advanced dysfunction. This is similar to the pattern seen with the progression of diastolic dysfunction and DT discussed earlier.²

Pulmonary vein inflow velocities measured during systole (S wave) and diastole (D wave) with pulsed-wave Doppler can also be used to determine the diastolic function. There are three components when measuring pulmonary inflow velocities: S wave (antegrade systolic wave), D wave (antegrade early-mid diastolic wave), and Ar (retrograde late diastolic wave that occurs due to atrial contraction). In normal diastolic function, the S wave velocity is larger than the D wave velocity with an S/D ratio >1 . As diastolic dysfunction develops and LV compliance worsens, there is a reversal of this ratio ($S/D < 1$) with the D wave velocity becoming larger than the S wave. As the LA pressure rises, the Ar wave becomes longer in duration as well. The best correlate with LVEDP on echo is the pulmonary venous flow reversal duration minus the mitral A wave velocity duration. The mitral A wave velocity represents the LA-LV pressure gradient that occurs with atrial contraction during late diastole. This is driven primarily by LV compliance and LA contractility. The less compliant the LV is at that point, the more there will be elevation in the LVEDP. However, there will also be more flow backward through the valveless pulmonary veins to accommodate the rise in pressure with increased LV stiffness.^{1,2}

THE HEMODYNAMICS OF DIASTOLIC FUNCTION IN HEALTH AND DISEASE STATES

NORMAL DIASTOLIC FUNCTION

Normal diastolic function can be seen on the hemodynamic LV tracing as a rapid fall in pressure, representing LV relaxation (Fig. 12.3). This begins with the closure of the aortic valve and continues until after the opening of the mitral valve. This downslope represents active LV relaxation, and the rate of deceleration is driven by the intrinsic elastic recoil and chamber compliance of the LV. The LV pressure tracing will continue to fall and exhibit a “dip” during early diastole. The nadir of this “dip” is the LV minimum pressure (LV min). A pressure gradient then exists between the LA and the LV minimum pressure, thus driving an efficient suctioning of blood flow across the mitral valve during early diastole once the mitral valve opens. This is seen more commonly among younger individuals whose ventricles can accommodate the rapid filling while the mean LA pressure remains normal. As diastole progresses and the LV fills, the LV pressure slowly rises and approaches the pressure of the LA. This point begins diastasis, and the rate of LV filling slows significantly. It is at this point where the mean LA pressure is recorded. The normal mean LA pressure is 6 to 10 mm Hg. With atrial contraction (“A” wave on the atrial and ventricular tracings), the LVEDP may rise to become even greater than the mean LA pressure, but it will still fall within the normal range. The LA-LV gradient generated by the A wave contributes to the final portion of LV filling at the end diastole. In those with normal diastolic function, the heart should be able to fill at consistently low pressures (mean LA pressure/LVEDP) both at rest and during exercise.^{1,2,4-6}

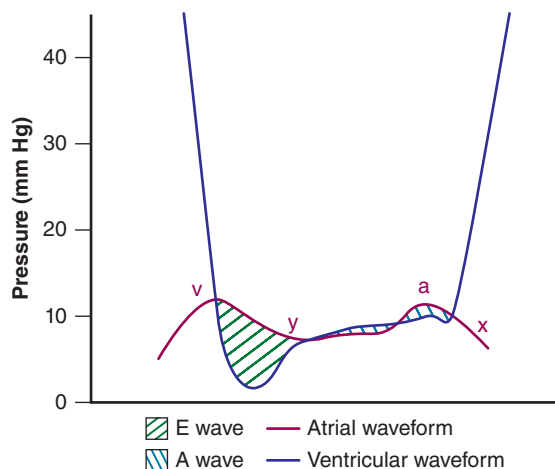


Fig. 12.3 Schematic representation of simultaneous left ventricular and left atrial pressures in a patient with normal diastolic function.

These findings are represented on echo as an E/A ratio between 0.9 and 1.5, an E/e' ratio of <14 , and normal e' tissue Doppler velocities at both the septal and lateral walls (Fig. 12.4). There is a subset of young, healthy patients with strong LV restoring forces and a hypercompliant efficient LV who display an E/A ratio of greater than 1.5. The high ratio is driven by a prominent E wave (mitral inflow velocity) and is due to an increased suctioning of blood from the LA to the LV once the mitral valve opens. They also display a large e', reflecting an increased intrinsic LV recoil (Figs. 12.5 and 12.6). On physical exam, this rapid filling of the LV will be associated with an auscultated S3 heart sound and is usually a normal finding in a young population without cardiomyopathy. In contrast, the S3 sound can also be auscultated in grade III/IV diastolic dysfunction, but, in these cases, it is due to an increased push of blood secondary to high LA pressures during early diastole, leading to a significant mitral inflow velocity.^{1,4,5}

ABNORMAL DIASTOLIC FUNCTION

Impaired Relaxation (Grade I Diastolic Dysfunction)

In the early stages of diastolic dysfunction, the main finding noted on the hemodynamic tracing of the LV is a delayed deceleration rate and a more gradual downslope on pressure waveform lasting until mid-diastole, which represents increased LV stiffness and worsening elastic recoil. This also results in a shallower LV minimum nadir point and, in turn, a narrower gradient between the LV and LA (Fig. 12.7). Compared with the normal diastolic function, the mitral inflow velocity (E wave on echo) will instead be reduced in the setting of this smaller gradient. During diastasis, LV filling slows as the LV pressure approaches the LA pressure; however, it is important to note that the mean LA pressure remains normal throughout this cycle. At the end of diastasis, a significant amount of the potential preload is still contained

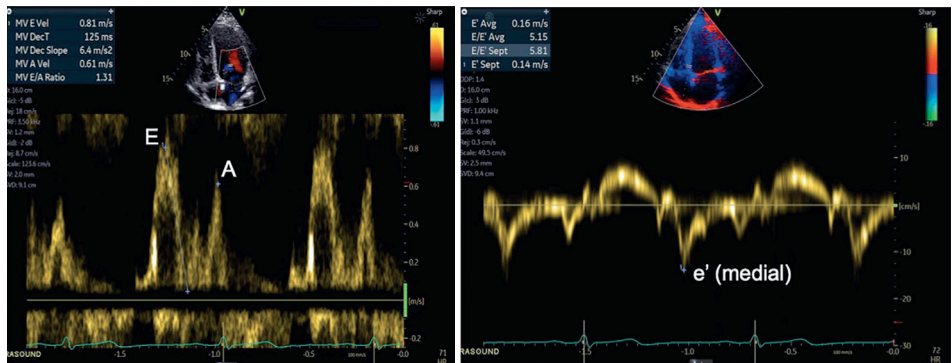


Fig. 12.4 Doppler echocardiography obtained in a patient with normal diastolic function demonstrating the normal E/A ratio (left panel) and medial tissue velocity (e') measured at the mitral annulus (right panel). MV, Mitral valve.

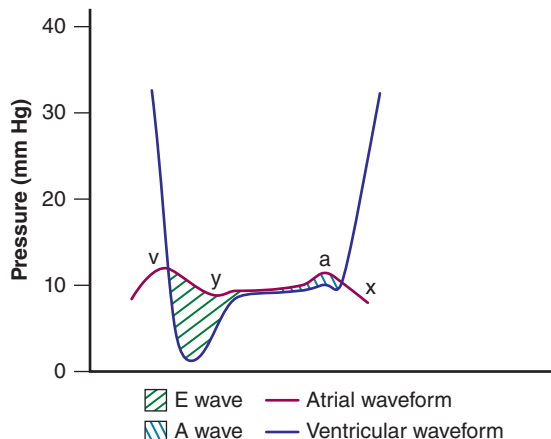


Fig. 12.5 Schematic representation of simultaneous left ventricular and left atrial pressures in a young healthy patient with a hypercompliant left ventricle and normal diastolic function.

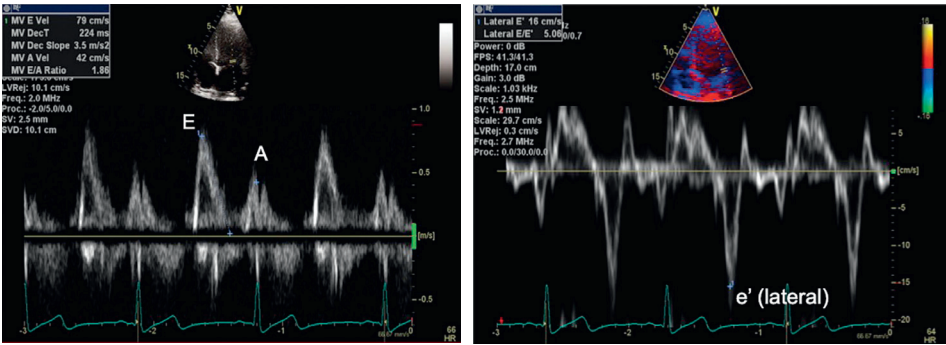


Fig. 12.6 Doppler echocardiography of an example of a young, healthy patient with a hypercompliant left ventricle, resulting in an elevated E velocity, but maintaining a normal E/A ratio (left panel) and normal e' values (right panel). MV, Mitral valve.

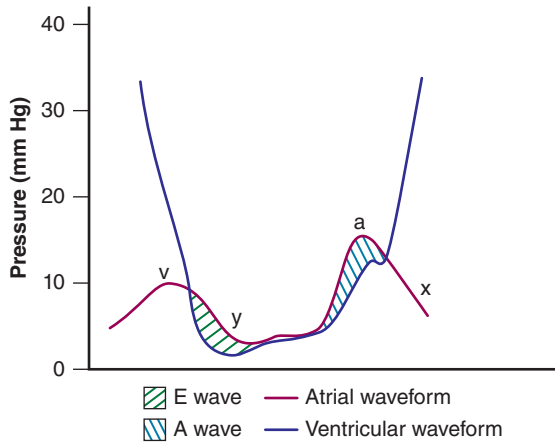


Fig. 12.7 Schematic representation of simultaneous left ventricular and left atrial pressures in a patient with grade I diastolic dysfunction (impaired relaxation).

within the LA. Applying the Frank-Starling law, an overstretched left atrium will result in a significant A wave on both atrial and ventricular tracings. This also results in a high LVEDP, which reflects decreased LV compliance (increased stiffness), and on a physical exam, correlates with the S4 sound. Of note, systolic dysfunction will always be associated with at least grade I diastolic dysfunction given its influence on elastic recoil and the LV deceleration rate.^{1,2,6}

On echocardiography, there will be an E/A ratio of <0.8 (in the setting of a reduced mitral inflow velocity and larger, more prominent A wave), and there will be a prolonged DT, a prolonged IVRT, and reduced e' tissue velocities (Fig. 12.8).²

Those with impaired relaxation rely on their atrial contraction to provide a significant proportion of LV filling and preload and maintain appropriate cardiac output. Symptoms such as dyspnea on exertion (DOE) are typically more likely to be due to an elevated mean LA pressure rather than an elevated LVEDP. Hence, these patients tend not to have symptoms at rest because of a normal mean LA pressure. However, on exertion, the mean LA pressure will rise, and their symptoms will become apparent. This clinical scenario of DOE in a patient with early diastolic dysfunction can be uncovered by the diastolic exercise stress test, which can be done invasively or noninvasively using a recumbent bike or repetitive hand grip.¹

Pseudonormalization With High-Filling Pressure (Grade II Diastolic Dysfunction)

As diastolic dysfunction becomes more abnormal, the LV diastolic pressure increases, and the LA pressure now has to overcome the higher LV pressure to maintain filling and will subsequently increase as well (Figs. 12.9–12.11). This causes the mitral valve to open earlier in diastole. This is apparent on pressure waveform tracings, where the V wave of the atrial waveform becomes more prominent and meets the LV deceleration slope at a higher point, resulting in a shorter IVRT. The deceleration rate (LV relaxation) continues to show an even slower and less steep slope. Grade II

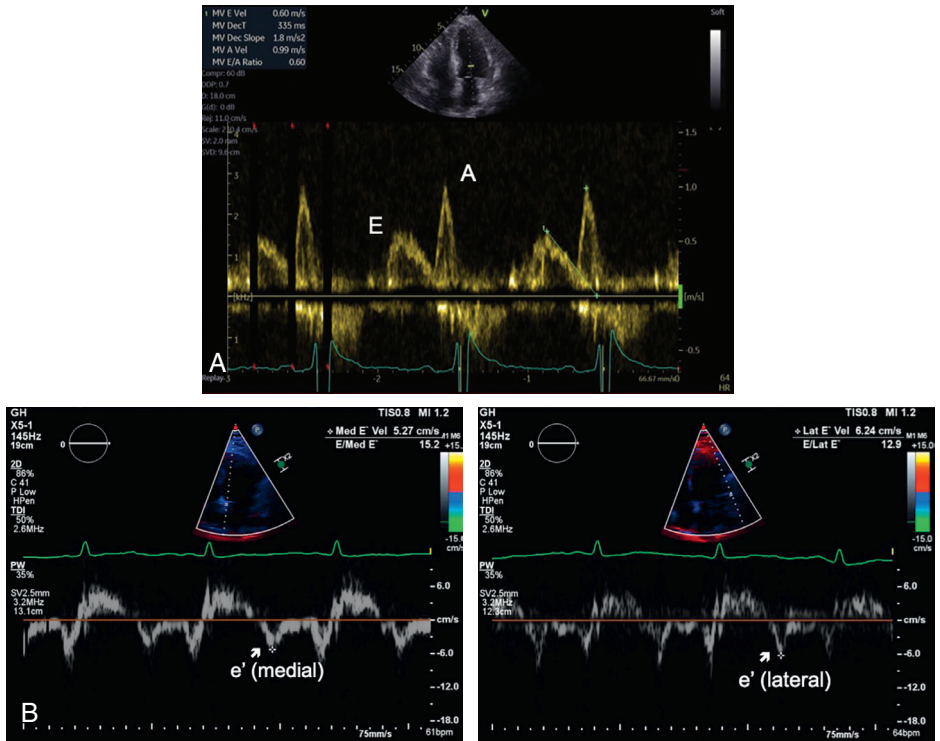


Fig. 12.8 (A) Doppler echocardiography demonstrating an elevated A wave relative to the E wave in a patient with grade I diastolic dysfunction (impaired LV relaxation). (B) Tissue velocities measured at the mitral annulus (represented as e') in a patient with impaired LV relaxation. Left panel is medial tissue velocities, and right panel is lateral tissue velocities. *MV*, Mitral valve; *PW*, pulsed wave; *TDI*, tissue Doppler imaging.

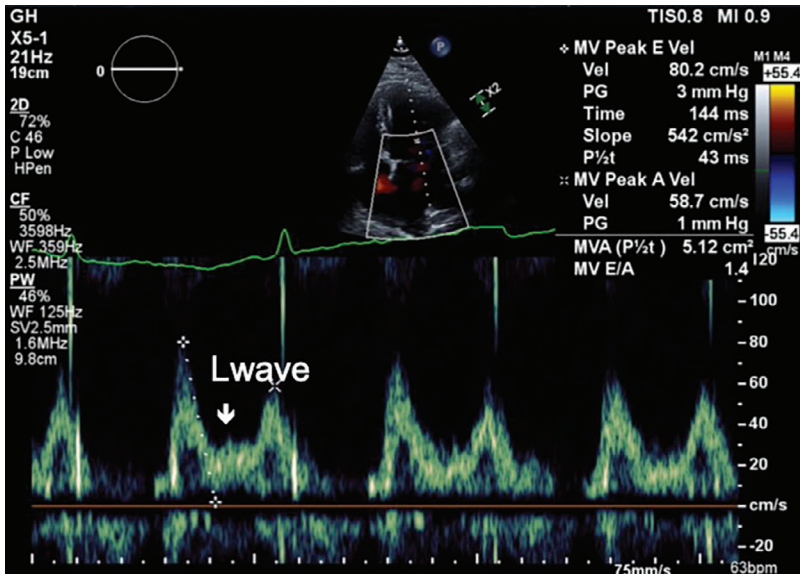


Fig. 12.9 Doppler echocardiography demonstrating an example of an “L wave” suggesting elevated left atrial pressure. *MV*, Mitral valve; *MVA*, mitral valve area; *PG*, pressure gradient; *PW*, pulsed wave.

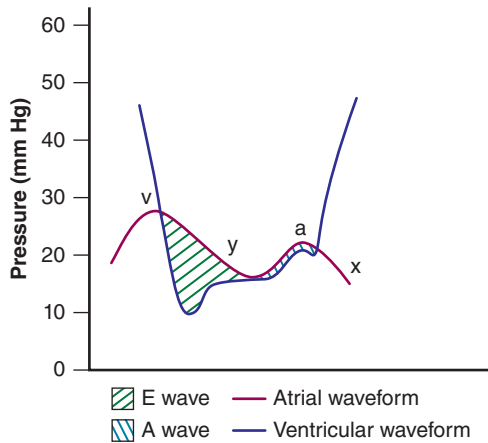


Fig. 12.10 Schematic representation of simultaneous left ventricular and left atrial pressures in a patient with grade II diastolic dysfunction.

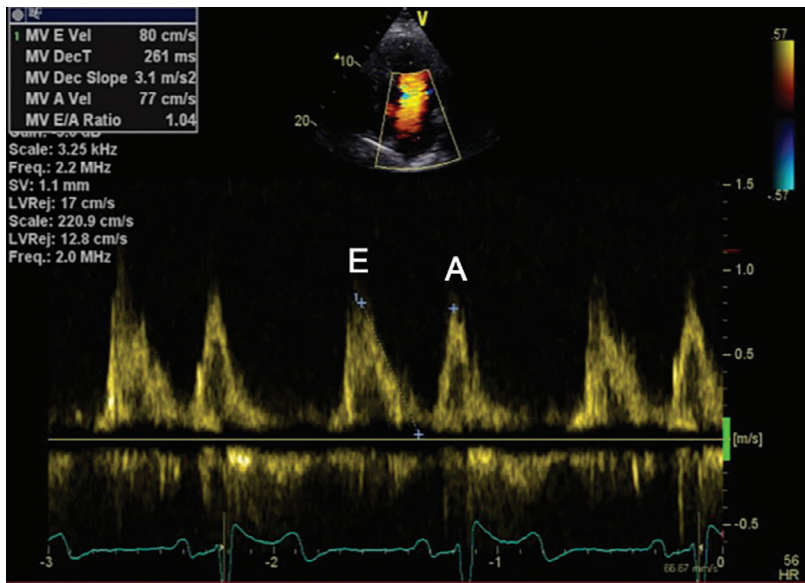


Fig. 12.11 Doppler echocardiography obtained in a patient with grade II diastolic dysfunction demonstrating a normal E/A ratio indicating pseudonormalization. MV, Mitral valve.

dysfunction is referred to as pseudonormalization because many of its parameters such as IVRT, DT, E wave, and E/A ratio will actually appear to be similar to those with normal diastolic function. The shorter IVRT is a result of the higher intersection point of the LA and LV pressure tracings despite the slower slope of deceleration of the LV waveform. As the E wave becomes large again (~50 cm/s or more) due to the now elevated LA pressure at rest, the E/A ratio will normalize from <0.8 (seen in grade I dysfunction) back to 0.8 to 1.5. The LV-LA gradient area will be very similar to that of normal diastolic function; however, the primary driving force behind flow from the LA to the LV in early diastole changes from the suction force developed by active LV relaxation in normal hearts to a pushing force from the significantly elevated LA pressure into a stiffer noncompliant LV. Parameters that remain abnormal include a lower e' velocities of both septal and lateral annular points given the less robust LV recoil, and therefore the E/e' ratio will now typically be elevated. With continued flow from the atrium to the ventricle during diastasis due to elevated LA pressure, an L wave

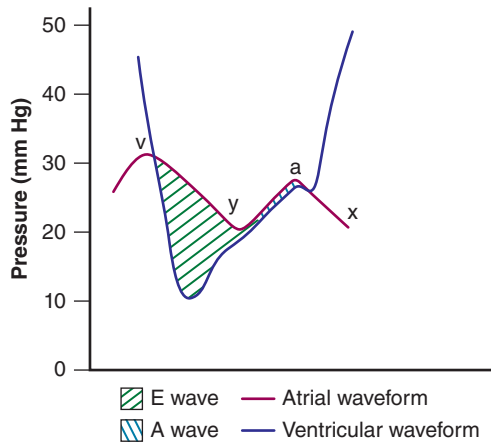


Fig. 12.12 Schematic representation of simultaneous left ventricular and left atrial pressures in a patient with grade III/IV diastolic dysfunction.

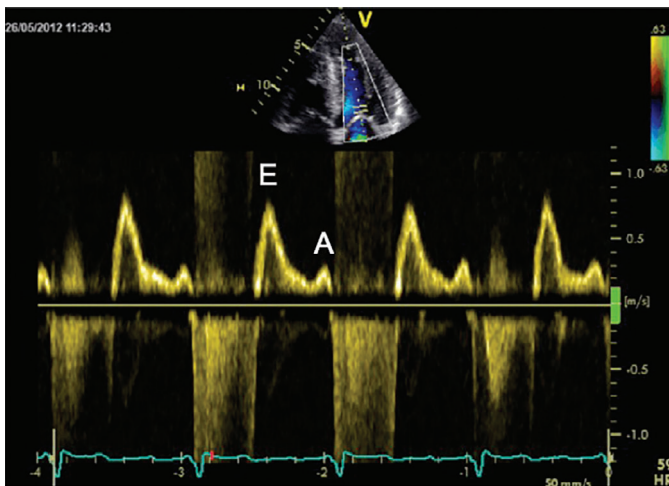


Fig. 12.13 Doppler echocardiography obtained in a patient with grade III/IV diastolic dysfunction demonstrating an abnormal and high E/A ratio consistent with a restrictive pattern.

can be seen on echo Doppler of the mitral inflow between the E wave and A wave (Fig 12.9). Since the mean LA pressure is high, and the LVEDP is even higher, this results in the likelihood of developing symptoms such as shortness of breath not only with exertion, but also at rest.¹

The Valsalva maneuver can be used to unmask this pseudonormalization pattern and reveal the relaxation abnormality. With Valsalva, LV preload is reduced. In those with normal diastolic function, there is a reduction of both E and A wave velocities to a similar extent, and as a result, the E/A ratio of normal individuals becomes unchanged with preload reduction. In those with a pseudonormalization pattern, the preload reduction will reduce the E wave velocity to a greater extent than the A wave velocity, lowering the E/A ratio to <0.8 and revealing what is typically observed during impaired relaxation.¹

Restrictive Pattern (Grades III/IV Diastolic Dysfunction)

When diastolic dysfunction becomes severe, the LV becomes even more noncompliant, and the LA pressure continues to rise (Figs. 12.12 and 12.13). This results in a restrictive pattern that may be reversible at first (grade III dysfunction), but can later progress to become irreversible (grade IV dysfunction). During early diastole, the active relaxation phase

and suction force from the LV have significantly diminished. The IVRT continues to shorten in the setting of elevating LA pressures, and the rate of deceleration (measurement of LV relaxation) continues to decline. With the LA pressure continuing to rise, there is a large LA-LV gradient that prematurely pushes the mitral valve open with rapid flow from the LA to LV. Given the worsened compliance of the LV, once the LV begins to fill, its pressure rapidly equilibrates to the mean LA pressure. This manifests as a prominent “dip” and a quick upward stroke on the LV pressure waveform and is identified as the “square root” sign. Given the rapid equilibration, there is very little contribution toward LV filling during early diastole and diastasis. Atrial contractility has also diminished and is unable to generate a significant amount of force to propel further flow into the already high LV filling pressures.

On echo this pattern will appear as a short IVRT of <70 ms, an E/A ratio of >2 (in the setting of a very large E wave and much smaller A wave), a DT of <140 ms due to the stiff LV, and even lower e' tissue velocities with a significantly larger E/e' ratio. If the restrictive pattern is still reversible (i.e., grade III dysfunction), then lowering the LA pressure by reducing preload may be able to improve the diastology to a pseudonormalization pattern (grade II dysfunction). The Valsalva maneuver can be used to determine if the restrictive pattern is reversible or not. With preload reduction, the E/A ratio >2 should decrease if there is reversibility. If the E/A ratio remains >2 with Valsalva, then the diastology has likely progressed to an irreversible restrictive pattern (grade IV dysfunction). The E/A ratio becomes fixed and unresponsive to other preload reducing maneuvers, such as diuresis. In these circumstances, the cardiac output of these patients is largely dependent on their heart rate (as the stroke volume becomes fixed), and bradycardia should be avoided.¹

Diastolic Dysfunction and Atrial Fibrillation

It is important to note that patients who display an impaired relaxation pattern only (grade I diastolic dysfunction) will depend largely on preload and atrial contraction to maintain adequate LV filling and cardiac output. When these patients are in atrial fibrillation, they lose their atrial contraction. Their LV filling becomes significantly less, and this results in not only a lower cardiac output but also increased LA pressures and worsening symptoms of shortness of breath secondary to increased pulmonary edema. In contrast, those who display a restrictive pattern (grade III/IV diastolic dysfunction) have progressed to the point where the amount of LV filling that occurs during atrial contraction is so small that the diastolic dysfunction is too far advanced for it to matter whether the A wave is present or not. Instead, this subset of patients typically depends more on the heart rate to maintain their cardiac output, as the stroke volume is already significantly impaired and fixed. If these patients were in atrial fibrillation, then lowering their heart rate excessively (i.e., strict rate control) would not be tolerated. Clearly, both groups of diastolic dysfunction patients, early and advanced, would certainly benefit from restoration of normal sinus rhythm.¹

Thus as diastolic dysfunction progresses and the LV becomes much stiffer, the LV tends to also become more volume sensitive. Symptoms and overall clinical status of the patient become much more tenuous and can change easily with small shifts in their volume status. This can be treated by preload reduction to lower the LA pressures or afterload reduction to increase the stroke volume and over time lower the LV filling pressures, which indirectly can lower the LA pressures. Diastolic dysfunction grades are best thought of as a pendulum of a clock that will shift from one side to another based on the fluid status, diuretics, exercise, and heart rate.¹

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